

WUJI Hand 2 on a Franka Panda: MuJoCo scenes for chopsticks and articulated tools, with contact validation

WUJI_Tools — simulation subsystem

October 5, 2026

Abstract

This document is the design record of the `WUJI_Tools` MuJoCo simulation: a 20-DoF WUJI Hand 2 (Beta 2, right hand) bolted onto a 7-DoF Franka Emika Panda, standing on a table, with one scene for chopsticks and one each for scissors, tweezers, pliers and a cordless drill, all with realistic objects to act on. We (i) convert the official WUJI description into a standalone, verified MJCF with precomputed convex-hull and CoACD collision geometry, (ii) identify the mount flange frame from the CAD mesh and compose arm and hand into one robot, (iii) solve a palm-up home configuration with a grid-seeded bounded least-squares inverse kinematics, (iv) design “sticky” chopsticks whose contact parameters are owned by the object through MuJoCo’s geom priority, (v) model food items (0.7 g to 8.4 g), rigid blocks, tableware, four articulated tools, a cuttable rope and card strip, nuts, a bolt and a wood screw on a helical joint coupling, and (vi) find a holding pose for two loose chopsticks and attempt the pick-up with the arm. Experiments show static scenes that settle within 0.5 mm, a thumb–index pinch that keeps a chopstick through a 180° wrist roll for $\mu \geq 0.6$, a 0.05 N to 0.2 N chopstick pinch that lifts every item, a two-chopstick holding pose that keeps both loose sticks within 7 mm in every gravity direction and closes the tips to 4 mm by index and middle flexion, the arm picking the 16 mm sugar cube up with the free sticks and a 71 mm supination lift (wider objects exceed this hold’s opening), scissors cutting the rope and the card at an opening of 0.28 rad, tweezers and pliers lifting a nut and a bolt, and the drill driving the screw 16.7 mm in 2 s against 16.8 mm expected from the pitch.

Keywords: dexterous hand, WUJI Hand 2, Franka Panda, MuJoCo, MJCF composition, inverse kinematics, contact friction, chopstick manipulation, tool use, cutting.

1 Overview

The repository provides eight scenes and a small Python package, `wuji_sim`, that wraps them:

- `scenes/panda_wuji_chopsticks.xml` (Fig. 1): Panda + WUJI Hand 2 on a 1.2×0.8 m table, hand palm-up 0.30 m above the table, two loose chopsticks under the hand, a plate with a tofu cube, a dumpling, a cherry tomato and a peanut, three rigid cubes (sugar, pine, a foam die) in a row, and an empty rice bowl.
- `scenes/panda_wuji_scissors.xml`, `panda_wuji_tweezers.xml`, `panda_wuji_pliers.xml`, `panda_wuji_drill.xml` (Fig. 4): the same arm and hand with one articulated tool under the hand and its task objects in front: a rope hanging from a hook stand and a card strip in a clip stand to cut; two M6 nuts; an M6 bolt and a nut; a clamped pine block with a started wood screw.

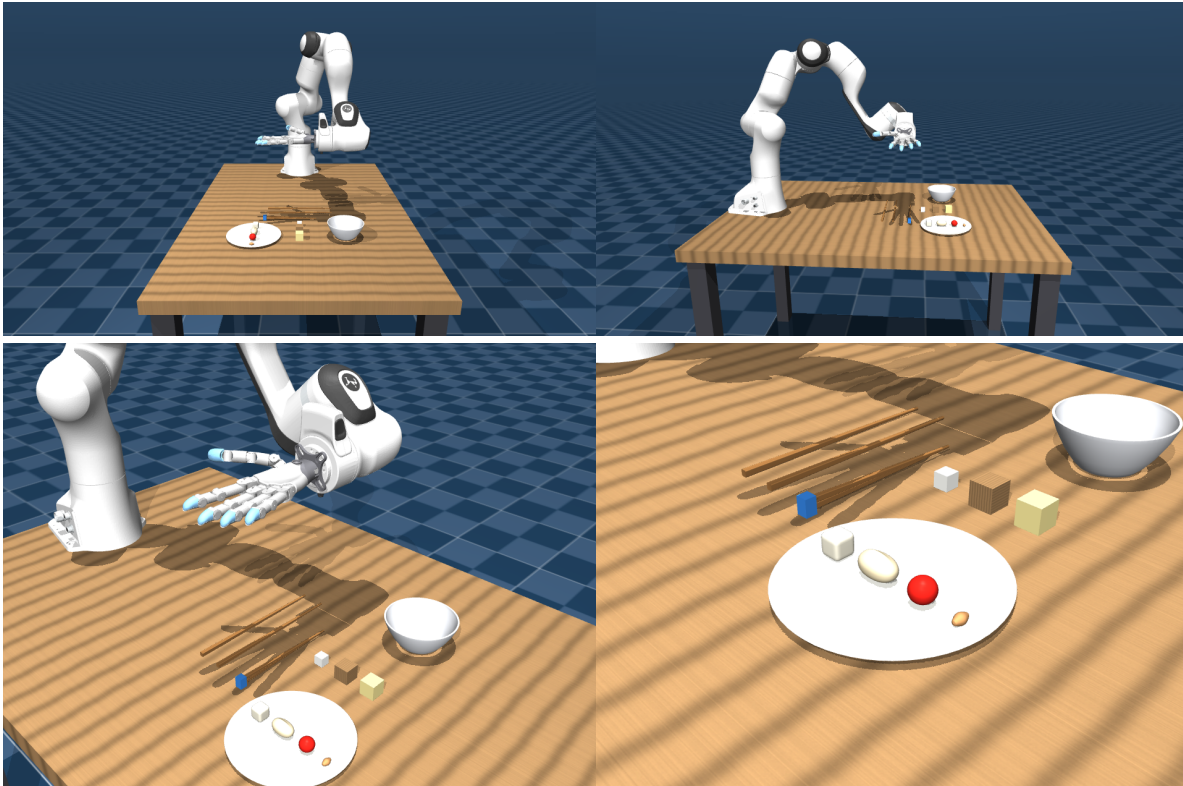


Figure 1: panda_wuji_chopsticks at keyframe palm_up: front and side views (top), the hand_closeup and plate cameras (bottom). The fingers point along $-y$ (to the robot’s right), the palm normal along $+z$; the chopsticks lie along y under the hand, the plate with food is to the right-front, the cubes straight ahead, the bowl to the left-front.

- scenes/wuji_hand2_right_chopstick.xml: the hand alone, fixed palm-up, with three chopsticks and a foam cube (grip search and finger-posture tuning).
- scenes/chopstick_pinch_test.xml and scenes/tool_bench_test.xml: mocap-driven chopsticks and tools plus the objects, benches that measure what the tools can do independently of the hand.

Every robot and object is a standalone MJCF in its own folder (robots/<name>/, objects/<name>/) with its meshes, a properties.yaml summary and its upstream licence; scenes assemble them with MuJoCo’s <attach> element. Section 2 documents the conversions, frames, objects and solvers; Section 3 the measurements; Section 4 how to run and extend the system.

2 Method

2.1 Hand model: provenance and conversion

The hand is the *WUJI Hand 2 Beta 2* right hand from wuji-technology/wuji-description (MIT licence; commit c2cd7f8, release 2026.9.11), file hand2/hand2_beta2/body/mjcf/right_with_mount.xml [3]. Beta 2 keeps the frozen Beta 1 coordinate contract (anatomical joint names, integer unit axes, r_wrist root) and adds one rigid tactile-pad link per fingertip. It is explicitly *not* the first-generation Wuji Hand that lives under hand/ upstream.

`scripts/tools/convert_upstream_hand.py` rewrites the upstream file into the single self-contained `robots/wuji_hand2/wuji_hand2_right.xml` (root body `r_mount`, the arm-flange mount) and copies the kinematic tree, inertials, joint limits, armatures, actuator gains and the ten contact exclusions *verbatim*; a regression check compares every joint, actuator and body of the generated model against the upstream model and reports zero differences, and the five fingertip sites coincide to 0 m. The only changes are structural:

- visual meshes in `meshes/right/visual/` (display group 2), collision meshes in `meshes/right/collision/` (group 3), the Menagerie convention; the five unused soft-pad meshes (`*_tip.STL`) are dropped;
- named default classes `wuji_hand2`, `wuji_hand2/visual`, `wuji_hand2/collision`, `wuji_hand2/pad` replace the upstream unnamed global default (required for the XML-level composition of Section 2.3);
- two extra sites: `flange` on the mount (the arm-side mounting frame) and `palm_center` on the wrist, whose x axis is the finger direction and z axis the palm normal.

The hand has 20 revolute joints, five fingers $\times$ (flexion, abduction, PIP/MCP, DIP/IP), total mass 0.6918 kg with the 69 g mount. Each joint is driven by a position servo $\tau = k_p(u - q) - k_v\dot{q}$ with $k_p \in [0.18, 0.69] \text{ N m rad}^{-1}$, $k_v \in [0.008, 0.031] \text{ N m s rad}^{-1}$ and force limits of 0.2 N m to 2 N m; upstream states that these gains are carried over from the first-generation hand pending system identification, so they are placeholders (Section 5).

Collision geometry. Upstream uses the raw link meshes and lets MuJoCo take their convex hull at load time. `scripts/tools/make_collision_meshes.py` precomputes the same hulls (qhull via `trimesh`) into explicit files, which keeps the collision shape inspectable and loads faster. For the palm shell (`r_wrist`) a single hull bloats the palm surface: measured from the palm-facing hull faces to the true mesh, the deviation is 1.81 mm mean and 6.19 mm maximum. We therefore replace it by an approximate convex decomposition with CoACD [4] (threshold 0.03, at most 12 hulls, resolution 2000, seed 0), which yields 12 pieces of total volume 105.3 cm^3 against 115.7 cm^3 for the hull, and reduces the palm-side deviation to 1.06 mm mean / 5.03 mm maximum. The decomposition takes 232 s and is committed; all other links (finger segments, tactile pads, mount) are thin shells for which a hull is the appropriate shape. The hand thus carries 38 collision geoms (12 palm pieces + mount + 25 finger links).

2.2 Mount frame identification

The with-mount model is rooted at `r_mount`, the arm-flange adapter, but upstream does not document which face mates with the arm. We analysed the mount STL directly (in its own frame, not MuJoCo’s recentred vertex buffer, which initially misled us): the mesh occupies $z \in [-41.8, 0] \text{ mm}$, its largest planar face is a $+z$ -facing disc of 1602 mm^2 at $z = 0$ with radius 28.5 mm, and the body is a tapered clamp below it; the wrist shell attaches at $z = -28.5 \text{ mm}$ and spans $z \in [-109, -21.5] \text{ mm}$. Hence the mount origin lies on the 60 mm arm-side flange plane with $+z$ pointing *into* the arm, the fingers extend along $-z$, and flexing all fingers by 0.8 rad moves the fingertips along (0.06, 0.81, 0.59), whose component orthogonal to the finger axis is $+y$: the palm normal is $+y$ and the thumb sits on $+x$ (a right hand, as expected).

The Panda’s tool frame `attachment_site` sits on the ISO 9409-1-A50 flange face with $+z$ pointing out of the flange. Mating the two planes therefore requires a half-turn about x ,

$$\mathbf{R}_{\text{site} \leftarrow \text{mount}} = \text{diag}(1, -1, -1) \quad (\text{quaternion } (0, 1, 0, 0)), \quad (1)$$

after which fingers point along the flange $+z$, the palm normal along the flange $-y$ and the thumb along the flange $+x$. The rotation about the flange axis is a free design choice (the bolt pattern allows 90° steps); Section 2.8 shows it does not affect reachability of palm-up poses.

2.3 Scene composition

MuJoCo ≥ 3.2 can attach a child model file under a body of the parent file, keeping the child’s own `meshdir/texturedir`, prefixing its names and namespacing its keyframes. We verified on MuJoCo 3.5.0 that this works for nested attachments, for several mesh directories, for `<include>` inside an attached body and for a free joint inside the child. It cannot, however, attach a child under a body that itself came from another attached child, so “hand on arm” needs one composite file. `scripts/tools/build_arm_hand.py` generates `robots/panda_wuji_hand2_right/` by merging the two component XMLs: because both only use named default classes, their `<default>`, `<asset>`, `<actuator>` and `<contact>` sections concatenate, the hand root is inserted into the Panda `attachment` body with Eq. (1), mesh paths are rewritten relative to `meshdir=“.”`, and the result is reloaded and compared with both components (joints, actuators, bodies, geoms: identical; flange-site offset 0 mm; finger direction $\cdot$ site $z = 0.989$; palm normal $\cdot$ site $y = -1.000$). An earlier attempt with the `MjSpec` API and `to_xml()` was abandoned because the exported XML nested the hand’s unnamed default under the Panda’s class (so hand joints would silently inherit the arm’s damping) and emitted bare mesh file names.

Scenes set the physics explicitly (child options are ignored under `attach`): time step 1 ms, `implicitfast` integrator, elliptic friction cones, `impratio=10`. The main scene has 49 bodies, 223 geoms, 132 meshes, $n_q = 104$ (7 arm + 20 hand + 11 free bodies), $n_v = 93$ and 27 actuators; names are prefixed `robot/`, `table/`, `chopstick_a/`, `plate/`, `tofu_cube/`, and so on.

2.4 Chopstick object

`scripts/tools/make_chopstick_mesh.py` generates a 240 mm stick whose cross-section morphs from a rounded 7.0 mm square (superellipse exponent 8) at the thick end to a 3.5 mm circle at the tip, linearly tapered. With bamboo density 650 kg m^{-3} MuJoCo’s exact mesh inertia gives $m = 4.21 \text{ g}$, centre of mass 28.2 mm from mid-length towards the thick end, and principal inertias $I_{\text{axial}} = 2.16 \times 10^{-8} \text{ kg m}^2$, $I_{\perp} = 1.74 \times 10^{-5} \text{ kg m}^2$. Contacts are carried by six tapered capsules (radius 3.6 mm to 2.0 mm), which give smooth rolling and several contact points along the finger links; the visual mesh carries the oak texture.

“Sticky” is implemented as a property of the object. The capsule class sets `priority="1"`, `condim="6"`, friction $(\mu, \mu_t, \mu_r) = (1.5, 0.02, 0.002)$, `solref=(0.005, 1)` and `solimp=(0.95, 0.99, 0.001)`. MuJoCo’s contact mixing rule uses the parameters of the higher-priority geom in full when priorities differ (otherwise the maximum friction and a weighted average of the impedance parameters), so every hand–stick, table–stick and stick–object contact uses these values and no hand or object parameter needs retuning; the hand keeps its upstream friction of 1.0 and the objects their own realistic values. The 6-dimensional cone adds torsional and rolling friction, which Section 3.4 shows to be decisive. Finally the free joint carries an armature of $1 \times 10^{-6} \text{ kg m}^2$: the stick’s axial inertia is only $2.16 \times 10^{-8} \text{ kg m}^2$, and a 3-dimensional pinch at 1 ms produced angular accelerations that MuJoCo flagged as unstable (peak angular velocity $1.2 \times 10^6 \text{ rad s}^{-1}$ before an automatic reset); with the armature the same trial peaks at 15 rad s^{-1} , while the nominal results are unchanged to the printed precision.

Table 1: Objects in the main scene. Mass is computed by MuJoCo from the collision primitive; friction is the (sliding, torsional, rolling) triple of the object’s own contact class, overridden by the chopstick’s values whenever a chopstick touches it.

object	size [mm]	ρ [kg/m ³]	mass [g]	collision	friction	solref [ms]
tofu cube	20 cube	1050	8.40	box	0.5, 0.005, 0.0001	10
dumpling	36 × 22 × 20	1000	8.29	ellipsoid	0.7, 0.01, 0.0002	10
cherry tomato	22 sphere	1000	5.58	sphere	0.6, 0.005, 0.001	10
peanut	15 × 9 × 9	1050	0.67	ellipsoid	0.8, 0.01, 0.001	5
sugar cube	16 cube	900	3.69	box	0.6, 0.01, 0.0001	5
pine cube	25 cube	450	7.03	box	0.45, 0.005, 0.0001	5
foam die	25 cube	120	1.88	box	0.9, 0.01, 0.0002	10
chopstick (×2)	240 long, 7→3.5	650	4.21	6 capsules	1.5, 0.02, 0.002 (prio.)	5
plate	180 dia, 8 high	–	160	disc + rim + feet	0.5, 0.005, 0.0001	5
bowl	120 dia, 50 high	–	130	disc + walls + feet	0.5, 0.005, 0.0001	5

2.5 Objects: food, blocks and tableware

Table 1 lists the objects. Food items are the targets of chopstick manipulation and are deliberately very light (real dimensions, bulk densities of the food), rigid blocks give flat faces that a two-point pinch holds geometrically, and the tableware is heavy enough not to be dragged. Each object is one MJCF: a visual mesh generated by `scripts/tools/make_object_meshes.py` (superellipsoids for the tofu and tomato, a waisted surface of revolution for the peanut, a pleated ellipsoid for the dumpling, surfaces of revolution for plate and bowl; all watertight) over a collision primitive that also defines the mass through its density, a per-object contact class, and a free joint with an armature of $1 \times 10^{-8} \text{ kg m}^2$ (at most $3 \times$ the peanut’s axial inertia). Compliance is modelled through the contact time constant `solref`: 5 ms for hard surfaces, 10 ms for tofu, dumpling and foam. Spheres and ellipsoids use `condim=6` with a rolling coefficient of 1×10^{-3} so they come to rest on the plate. The plate is a 2.5 mm floor disc with sixteen rim blocks and an eight-sphere *foot ring*; the bowl a base disc, foot ring and 32 tilted wall blocks in two rings. The foot rings are not decoration: a cylinder resting on a box produces a single contact point in MuJoCo, and the plate then rocked enough to roll the tomato 20 mm during settling; with eight contacts the drift is 0.05 mm (Section 3.1). The table top and tableware use a stiff, high-impedance contact (`solref` 5 ms, `solimp` 0.95–0.99) so the 160 g plate sinks 0.01 mm; a procedural oak texture is applied to the table, the chopsticks and the pine cube.

2.6 Articulated tools and their task objects

Four tools with real joints live in `objects/tools/`, built from primitives with the dimensions and masses of real items (Table 2). Each is one body (half A) with the moving part as a child body on a hinge; the tool files are passive, and a scene that wants to drive a tool adds an actuator on the attached joint name (the bench does; the drill carries its own motor). The scissors’ two halves cross at the pivot like a real lap joint (blades stacked in z , cutting edges at $y = 0$). The tweezers’ hinge carries a flexure spring $\tau = k(q - q_0)$ with $k = 1.0 \text{ N m rad}^{-1}$ and $q_0 = 0.14 \text{ rad}$, i.e. a 15 mm tip gap at rest and 1.3 N at the tips to close. The drill is a pistol-grip cordless driver standing on its battery: a spring-return trigger (slide, 6 mm, 500 N m^{-1}), a chuck hinge with the motor-plus-gearbox reflected inertia of $4 \times 10^{-4} \text{ kg m}^2$ as armature, and a velocity servo ($0.2 \text{ N m s rad}^{-1}$, $\pm 3 \text{ N m}$, up to 60 rad s^{-1}); with a $20 \times$ smaller armature and a $2.5 \times$ larger gain the discrete servo was unstable and ran away to -61 rad s^{-1} , a time-constant below the time step.

Table 2: Articulated tools (all primitives). Mass is the compiled model mass; the working surfaces are the blades, pads, jaw tips and bit.

tool	size	mass	joints, passive dynamics, working-surface contact
scissors	175 mm; blades 70×12×1.5 mm	33 g	pivot 0 (closed) to 0.7 rad, damping 0.01, friction loss 0.02 N m; steel $\mu = 0.4$
tweezers	120 mm; arms 1×8 mm	20 g	pivot 0 to 0.22 rad, spring 1.0 N m/rad with rest 0.14 rad; pads $\mu = 0.9$, priority 1
pliers	165 mm; needle-nose jaws 60 mm	165 g	pivot 0 to 0.8 rad, damping 0.02, friction loss 0.05 N m; tip pads $\mu = 1.0$, priority 1
drill	160 mm tall, pistol grip	435 g	trigger slide 6 mm with 500 N/m spring; chuck hinge, armature 4e-4 kg m ² , velocity motor 0–60 rad/s, ± 3 N m; bit $\mu = 0.4$

Cuttable rope and card. MuJoCo cannot cut geometry, so a cuttable object is a chain of free segments joined by `connect` equality constraints, and a cut deactivates a constraint at run time (`data.eq_active`). The rope is ten 20 mm capsule segments of 8 mm cotton rope (0.75 g each, neighbour contacts excluded) hung from the hook of a desk stand by one more connect; the card strip is six 25 mm pieces of 160 g m⁻² card (0.12 g, 0.8 mm contact thickness) with two connects per joint at the strip edges, welded at the top into the jaws of a clip stand. Because a child model cannot carry several free bodies through one `<attach>`, the chains are composed in the scene files from single-segment objects (`objects/cuttables/`). The cut model (`wuji_sim.tasks.cut_update`) is geometric: in the frame of half A (pivot at the origin, blade along $+x$, A’s edge at $y = 0$, B’s edge at $y = -x \sin \theta$), a segment at (x, y) with $-(x \sin \theta + 6 \text{ mm}) \leq y \leq 6 \text{ mm}$ is cut when

$$\theta \leq \frac{w}{x}, \quad w = 10 \text{ mm}, \quad (2)$$

i.e. when the crossing point of the two cutting edges, which travels outwards as the blades close, reaches it; the link below the segment is deactivated and the freed piece falls.

Hardware. `objects/hardware/` holds an M6 nut (hexagonal prism, 2.4 g; the flats matter: with a cylinder collision the tweezer pads could not hold it), an M6×30 bolt (9 g; with a 4-dimensional cone it rolled off across the table on its shank at 35 mm s⁻¹ after settling, so it carries rolling friction 1×10^{-3} like the tomato) and a clamped pine block with a #8 wood screw started 12 mm into a pilot hole. The screw has a slide joint (depth s) and a spin joint (ϕ) coupled by an equality

$$s = \frac{p}{2\pi} \phi, \quad p = 1.8 \text{ mm per turn}, \quad (3)$$

with 0.15 N m of friction loss on ϕ for the wood; the shank starts inside the block, so the block–screw contact is excluded (without it the contact solver pushed the screw 12 mm out of the hole). Bit engagement is emulated: when the bit tip is within 2 mm of the screw head the bench activates a second equality $\phi = -\theta_{\text{chuck}}$ (the drill points down, so positive chuck rotation is clockwise seen from above), so the screw turns with the chuck and Eq. (3) drives it in while the drill descends at $p\omega/2\pi$.

2.7 Two-chopstick grip and pickup

The two chopsticks are two free bodies (`objects/chopstick/chopstick.xml`, 240 mm, 4.21 g); nothing is welded or strapped, the sticks are held by contact and friction alone. Human chopstick use is

not one grip: Yang *et al.* [8] learn skills for many holding styles found by Bayesian optimisation over a physics check, and the Marcosticks taxonomy [10] classifies more than forty grips by which fingers hold the top and the bottom stick in the open and closed postures. The textbook standard grip (lower stick fixed in the thumb web and on the ring finger, upper stick held like a pencil by thumb, index and middle pads) and the whole lateral family rest the sticks’ rear ends on the purlicue, the compliant web between thumb and index. This hand has none: its thumb metacarpal (`r_thumb_proximal_abd`) lies on the radial edge of the palm 4 mm to 35 mm proud of it (palm frame $x \in [-14, 31]$, $y \in [-53, -26]$ mm), a curled finger ($\text{MCP} \geq 0.8$, $\text{PIP} \geq 1.0$ rad) is a closed loop whose middle phalanx walls off the pad, and the finger lanes are 20 mm wide with 4 mm gaps (index $y \in [-43, -23]$, middle $[-19, -2]$, ring $[0, 20]$, little $[17, 39]$ mm when extended, at $z \in [-21, -3]$ mm). The grip used therefore combines the two human mechanisms that need no purlicue: the *upper* stick is sandwiched between the index and middle fingers and swung by their MCP flexion (the Dangling Stick grip of [10], “the knuckles do most of the work”), and the *lower* stick is sandwiched between the ring and little fingers on the ulnar side (the Forsaken Pinky grip with the ring finger as the second jaw). Each stick is clamped near its midpoint, so gravity produces no pitch torque about the clamp, and the two finger pairs converge (index/middle abducted towards the ring, ring/little towards the middle) so that the sticks, which align with their pair, meet at the tips.

Holding-pose search. A holding pose is the 20 joint angles plus one line per stick in the palm frame, each through two points at the stations $X_0 = 90$ mm (clamp) and $X_1 = 210$ mm (near the tips): $\mathbf{d} = (\mathbf{p}_1 - \mathbf{p}_0)/\|\mathbf{p}_1 - \mathbf{p}_0\|$, stick centre $\mathbf{c} = \mathbf{p}_0 + (L/2 - B)\mathbf{d}$ with $B = 110$ mm from the thick end, orientation $[R\mathbf{d}, \mathbf{y}, \mathbf{z}]$ with $\mathbf{y} = \mathbf{n} \times R\mathbf{d}$ and $\mathbf{z} = R\mathbf{d} \times \mathbf{y}$ ($R, \mathbf{n}$ the palm rotation and normal). `scripts/tools/search_chopstick_grip.py` samples the 28 parameters uniformly in a box (listed in the script) subject to: each pair squeezes (index minus middle abduction ≥ 0.12 rad, ring minus little ≥ 0.12), the lines meet within 30° , the extrapolated tip separation is 10 mm to 35 mm and the upper tip starts below the lower one. Stage 1 (geometry): the clamping joints are backed off (thumb abduction -0.3 , index and ring -0.3 , middle and little $+0.3$ rad), each stick is teleported onto its line and lifted along the palm normal in 1 mm steps until no hull is penetrated by more than 0.5 mm (at most 30 mm), the targets ramp to the grip posture over 0.4 s, and the sample is kept only if the lower stick touches two links of the ring/little pair and none of the index or middle and the upper stick touches an index and a middle link. Stage 2 (dynamics): with gravity set in turn to palm-up, tips-down, little-finger-down and thumb-down (the orientations the pick-up passes through) and then palm-down and tips-up, the hand settles 0.5 s each and the drift $\delta_k = \max(\|\mathbf{c}_L - \mathbf{c}_L^0\|, \|\mathbf{c}_U - \mathbf{c}_U^0\|)$ must stay below 10 mm for the four task directions; the index and middle MCP targets then ramp by up to 0.6 rad (half of it at the PIP) over 0.8 s while the minimum tip gap g^* , the lower-stick motion Λ at that moment and the reopened gap g_{re} are recorded. The score is

$$J = -g^* - 2\delta_{\text{max}} - 2\Lambda - 0.3|g_{\text{re}} - g_0| - 0.5\max(0, g_0 - 40\text{ mm}) - 0.5\max(0, 15\text{ mm} - g_0), \quad (4)$$

with g_0 the open gap. A stick between two finger sides is free to yaw about the palm normal and aligns with its pair, so with pair axes converging by α and clamp stations Δy apart at x_c the sticks meet at $x_{\text{meet}} = x_c + \Delta y/(2 \tan(\alpha/2))$, placed at the tips by $\alpha \approx 0.4$ rad.

Result. The pose the search settled on (`grip_search_results.json`; 28 parameters, found by the insert-and-close search over 2500 geometric samples and 120 dynamic evaluations, then re-scored with the closed-loop pinch) holds the sticks with the fingers nearly straight: the lower stick between

Table 3: Two-chopstick holding pose, hand only: drift of the stick centres after 0.5 s in each gravity direction (palm frame), and the closed-loop pinch for three finger assignments (axis gap near the tips: open, minimum reached, after reopening).

gravity direction	palm up	tips down	little f. down	thumb down	palm down	tips up
lower stick drift [mm]	0.5	2.6	3.6	4.4	6.7	5.4
upper stick drift [mm]	0.6	1.5	2.6	3.1	5.6	4.2
pinch	open [mm]	min [mm]	flexion [rad]	held [mm]	reopen [mm]	lower moved
index + middle	41.5	4.1	0.19	25.4	45.2	1.0
index only	41.5	13.5	0.45	12.2	53.6	1.2
index + thumb press	41.5	20.3	0.54	8.4	35.2	1.2

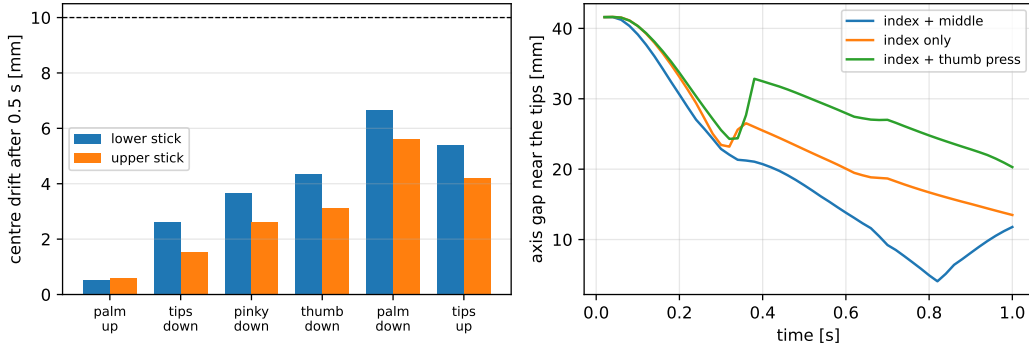


Figure 2: Hand-only holding pose: stick-centre drift under six gravity directions (left, dashed line = the 10 mm tolerance) and the tip-region axis gap during the closed-loop pinch for the three finger assignments (right).

the middle phalanges of the ring and little fingers with its rear end on the palm shell, the upper stick between the middle phalanges of the index and middle fingers with the thumb’s proximal phalanx lying on its rear end (contacts: lower ring_middle, ring_distal, pinky_middle, wrist; upper index_middle, middle_middle, ring_distal). The contact forces on the sticks are 8.5 N (lower) and 4.1 N (upper). Table 3 and Fig. 2 give the measurements of `exp_chopstick_grip.py`: the stick centres drift by at most 4.4 mm under the four task orientations and 6.7 mm with the palm down, so the hold is not orientation-dependent; the closed-loop pinch with index and middle closes the tip-region axis gap from 41.5 mm to 4.1 mm with the lower stick moving 1.0 mm, the index alone reaches 13.5 mm, and adding a thumb press (abduction +0.25 rad) reaches only 20.3 mm because on this hand the thumb lies on the stick’s rear end and pressing it translates the stick rather than levering its tip. Humans close the standard grip with index and middle together [10], which is what the controller does; the thumb’s job here is to hold the rear end, not to close. The actuator gains used for these numbers are the upstream placeholders scaled by `config.HAND\_GAIN\_SCALE` ($\times 3$ on the abduction servos, $\times 1.5$ on flexion): with the upstream values the abduction servos (0.2 N m) let the sticks shift 5 mm to 9 mm per orientation change and the arm’s accelerations displaced them by tens of millimetres.

What was tried first. (i) A random 28-dimensional search with the sticks along the fingers and strict insertion into the static posture found 0 of 3400 samples feasible: the feasible set is a thin shell. (ii) A constructed standard grip (lower stick under the thumb’s proximal phalanx on

the index knuckle, upper stick on the thumb under the index pad) had no clearance or dropped the lower stick: nothing supports a stick outboard of the palm ($y < -41$ mm) and the metacarpal blocks it behind $x = 31$ mm. (iii) Lateral finger-pair clamps behind the stick midpoint held in all six directions for 22 of 80 samples, but the tip-heavy sticks pitched about the clamp and dangled through the palm plane (tip gaps 90 mm to 280 mm). (iv) Midpoint clamps with nearly straight fingers and parallel pairs held with 6 mm to 8 mm drift, but the sticks aligned with the parallel pairs and the tips stayed 88 mm to 108 mm apart whatever the fingers did. The converging pairs of the final grip follow from (iv).

Pick-up sequence. `scripts/tools/exp_chopstick_pickup.py` (and the viewer demo `scripts/demo_chopsticks_pickup.py`) reuses the holding pose in the arm scene. The sequence was rebuilt around measured quantities after every scripted assumption about the in-hand geometry failed (Section 3.6):

1. *Verified take.* The sticks are inserted and the fingers close as in the search; the pair squeeze is then increased by 0.1 rad (index towards the middle, middle towards the index). The thumb is *not* pressed onto the upper stick: a thumb press pins the stick’s rear end and removes the index/middle closing degree of freedom. The in-hand geometry is measured (contacts, closing direction, axial tip offset, gap) and the take is repeated with perturbed insertion lines until it passes.
2. *Measured Jacobian.* The upper tip’s response to +0.06 rad of index+middle MCP flexion (PIP at half rate) and of their common abduction is measured in the palm frame, giving a 3×2 Jacobian J that the pinch controller inverts by least squares; for this pose $J \approx [(-34, 14, -37)^\top, (-77, 58, -100)^\top]$ mm/rad.
3. *Closing direction and orientation.* The closing direction $\mathbf{c}$ is the unit vector from the upper tip to the closest point of the lower stick’s axis (the “closing point”); the hand orientation is chosen on a grid of stick tilts (10° – 60° below horizontal) and rolls about the stick axis to maximise the palm’s upward component subject to $\mathbf{c}$ being horizontal (between 20° downward and 3° upward). The hold is strongest palm-up (Table 3), and the pinch must sweep across the object, not scoop it.
4. *Placement.* The closing point is placed at the object’s far face along $\mathbf{c}$ ($w/2 + r + 3$ mm from the centre, w the object width, r the stick radius) with the lower tip 3 mm below the object centre and never below the object’s support, 60 mm above it first, then down over 2 s, then up to six closed-loop corrections of the closing point on the commanded palm pose (the position servos sag). Objects wider than the natural separation at the tips use a pinch station further back along the sticks where the separation is sufficient; opening the sticks by more than about 35 mm with the fingers lets the sandwich go.
5. *Pinch.* Every 20 ms the controller moves the upper tip towards the closing point by a least-squares step on J (at most 0.015 rad per step) and stops when both sticks press the object with more than 0.3 N, then adds a 0.04 rad squeeze.
6. *Supination lift.* The palm rotates about the horizontal axis $\mathbf{n} \times \hat{\mathbf{z}}$ until its normal $\mathbf{n}$ points up, in 40 damped-least-squares IK steps, while rising 50 mm: the hand turns from palm-sideways to palm-up and the tips swing the object up, as a person turns the wrist.

2.8 Palm-up home configuration

Let $\mathbf{p}(\mathbf{q})$, $\mathbf{R}(\mathbf{q})$ be the position and rotation of the `palm_center` site for arm joints $\mathbf{q} \in \mathbb{R}^7$ (hand joints at zero). The target is $\mathbf{p}^* = (0.10, 0, 1.05)$ in the world (0.30 m above the table, 0.55 m in front of the base) and $\mathbf{R}^*$ with palm normal $+z$ and fingers along $-y$. We first implemented damped least

squares [5] with the residual $\mathbf{e} = (\mathbf{p}^* - \mathbf{p}, \log(\mathbf{R}^* \mathbf{R}^\top))$, $\Delta \mathbf{q} = \mathbf{J}^\top (\mathbf{J} \mathbf{J}^\top + \lambda^2 \mathbf{I})^{-1} \mathbf{e}$ and joint-limit clamping. It converges in 11 iterations on a nearby target but is trapped at the joint-6 limit from the Panda home pose for the full rotation, so the production solver (`PandaArm.ik_site_global`) scores a cached 7^7 joint grid by $10 \|\mathbf{p}^* - \mathbf{p}\| + \|\mathbf{R}^* - \mathbf{R}\|_F$, refines the 15 best seeds with bounded least squares (`scipy.optimize.least_squares`, joint limits as bounds) and ranks the converged solutions by joint-limit margin $\min_i \min(q_i - q_i^{\min}, q_i^{\max} - q_i)$. The DLS step is kept to polish the chosen solution.

A kinematic fact drove the choice of $\mathbf{R}^*$. The fingers lie along the flange axis, so “palm up with fingers straight ahead” at table height forces the flange to point forward and low; a 9^7 grid over the whole joint space (4 782 969 configurations, 44s) found 1338 palm-up poses in the box $x \in [0.35, 0.70]$, $|y| < 0.12$, $z \in [0.15, 0.45]$ relative to the base, but every solution with the fingers within 10° of $+x$ had zero margin (joint 5 or 6 pinned), while poses with the fingers roughly sideways had margins above 1 rad; a sagittal-plane search with any of the four 90° mount angles found no straight-ahead solution either. Pointing the fingers along $-y$ (a “tray” pose, thumb towards the robot) gives twelve exact solutions for the target above, the best with margin 1.035 rad:

$$\mathbf{q}_{\text{palm_up}} = (0.7558, 0.4352, -0.4802, -2.0363, -1.2071, 1.5081, 1.6224) \text{ rad}, \quad (5)$$

position and orientation residuals $< 10^{-6}$, no robot-table or self contacts. `scripts/tools/solve_palm_up_pose.py` writes it into `scenes/keyframes/`, taking every object’s pose from the compiled scene.

2.9 Grasp posture

For the hold experiment a scripted grasp was tuned in the hand-only scene (`scripts/tools/tune_grasp_posture.py`): a chopstick is laid across the palm at offset x_o along the fingers (lifted along the palm normal in 1 mm steps until it no longer intersects the hand, a necessity we learned when a 9.6 mm teleport penetration launched the 4 g stick at 170 ms^{-1}), the fingers settle open for 0.8 s, then ramp to the posture over 1 s. The adopted posture, `config.HAND_GRASP_CHOPSTICK`, flexes the thumb (cmc flexion 1.1, abduction $+0.6$, mcp 1.0, ip 0.9 rad) over the index base and curls the four fingers (mcp 1.2, pip 1.3, dip 0.8 rad); at $x_o = 0.075 \text{ m}$ it pinches the stick between the thumb pad and the index proximal link (Table 5).

3 Experiments

All experiments are deterministic scripts in `scripts/tools/` that print their tables and write vector figures to `docs/figures/`; `run_all_experiments.py` regenerates everything in dependency order. Timings are single-threaded on an Apple-silicon laptop, MuJoCo 3.5.0.

3.1 Static validation and throughput

`exp_table_settle.py` simulates every arm scene for 3 s from `palm_up` with the actuator targets equal to the keyframe (Fig. 3, Table 4). Every free body stays where the scene puts it: the chopsticks settle by 0.72 mm as the thinner tip comes to rest, the tofu and the peanut by 0.47 mm, the cubes and the tableware by less than 0.03 mm; in the tool scenes the tools settle by 0.02 mm to 0.2 mm, the nuts by 0.9 mm, the rope segments by 3 mm to 4 mm (the hanging chain straightening) and the card pieces by $\approx 0.4 \text{ mm}$. The deepest contact penetration is 0.5 mm for the food and blocks and $\approx 0.5 \text{ mm}$ in the tool scenes, after two fixes this test exposed: the clipped top card piece and the knotted top rope segment overlap their holders and have those contacts excluded. The arm

Table 4: Settle test: displacement of free bodies from their keyframe pose over 3 s (chopstick scene).

body	mass [g]	max disp. [mm]	final [mm]	max rotation [deg]
chopstick a / b	4.21	0.722	0.625	0.49
plate	160	0.002	0.002	0.00
tofu cube	8.40	0.470	0.413	2.26
dumpling	8.29	0.050	0.050	0.04
cherry tomato	5.58	0.050	0.049	0.00
peanut	0.67	0.478	0.478	0.00
sugar / pine / foam cube	3.69 / 7.03 / 1.88	≤ 0.03	≤ 0.007	0.00
bowl	130	0.002	0.002	0.00

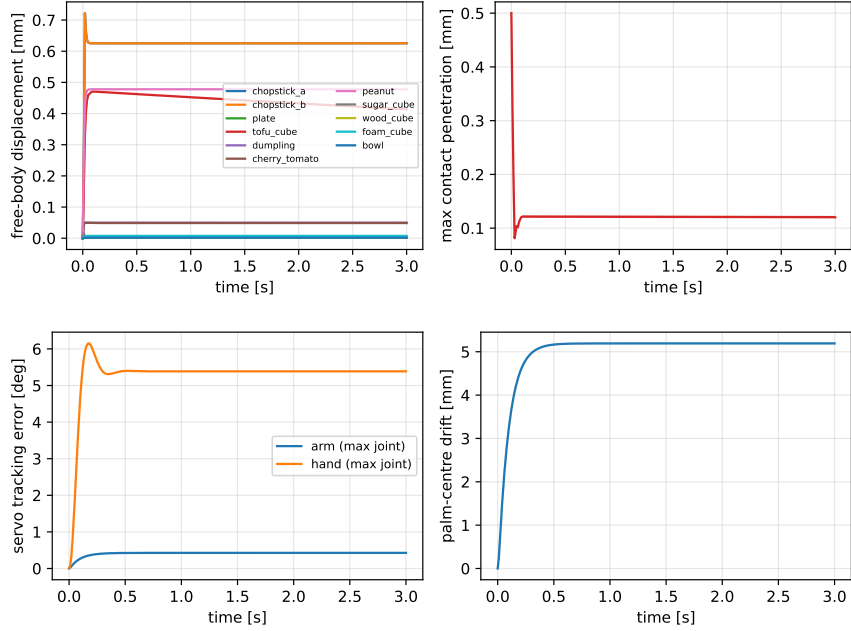


Figure 3: Static validation over 3 s (chopstick scene): free-body displacement and deepest contact penetration (top), servo tracking error of arm and hand and palm-centre drift (bottom).

holds the pose with a maximum joint error of 0.43° , which moves the palm centre by 5.2 mm (the Menagerie position servos are finite-stiffness, $k_p = 2000\text{--}4500$); the hand joints sag by 6.15° peak / 5.39° final under gravity with the palm up (the MCP flexion joints hyper-extend by $4.4^\circ\text{--}5.4^\circ$), a direct consequence of the placeholder gains. Throughput (single thread) is 1870 steps per second ($1.9\times$ real time) for the chopstick scene with 223 geoms, 2780 ($2.8\times$) for the scissors scene with its 147 DoF chains, 5750–5980 ($5.7\text{--}6.0\times$) for tweezers and pliers, and 10 400 ($10.4\times$) for the drill scene.

3.2 Palm-up inverse kinematics

The global solver returns 12 distinct solutions with margins from 1.035 rad down to 0.001 rad; all are collision-free. Fig. 5 shows the DLS polish converging from perturbed seeds ($\sigma = 0.1\text{--}0.5$ rad) in a few iterations, and stalling from the Panda home pose, which is why DLS alone was insufficient.

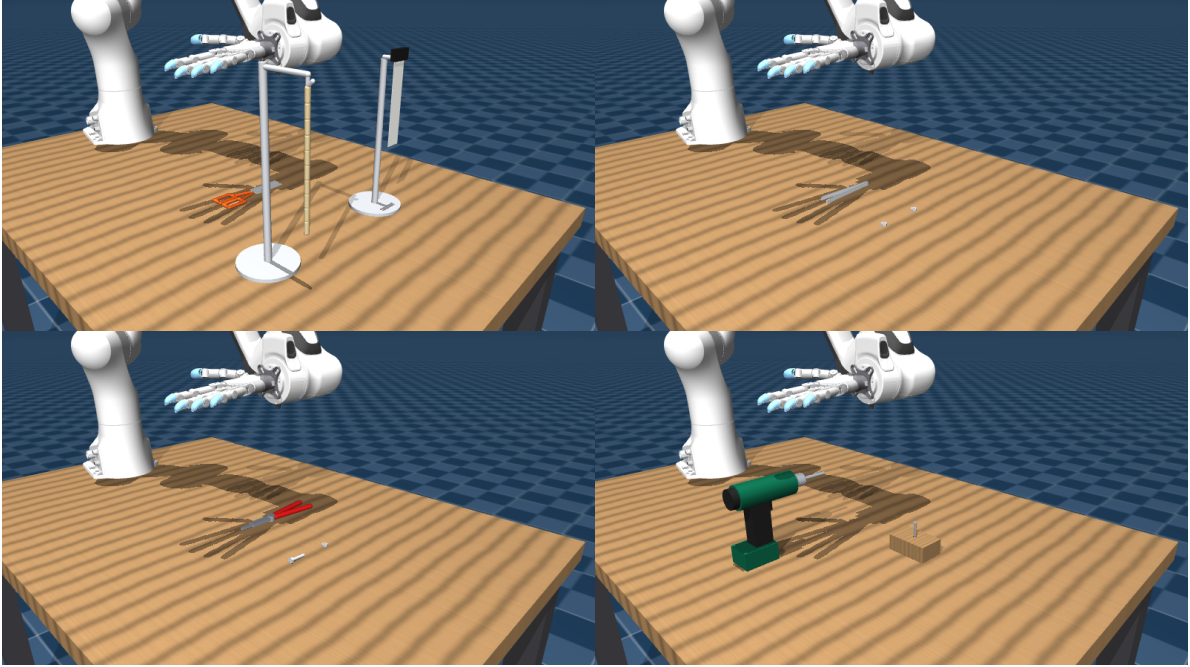


Figure 4: The four tool scenes at keyframe `palm_up` (workspace camera): scissors with the rope on the hook stand and the card in the clip stand; tweezers with two M6 nuts; needle-nose pliers with an M6 bolt and a nut; the cordless drill standing on its battery and the clamped pine block with a started screw.

3.3 Grasp posture and placement sweep

Table 5 summarises three postures $\times$ four placements. Only the stick lying across the proximal phalanges ($x_o = 0.075$ m) produces a loaded grasp, a two-contact thumb-index pinch of 2.9 N to 4.8 N; closer to the palm the fingers merely cage the stick ($j0.3$ N), further out they miss it. More thumb flexion (`B_thumb`) raises the pinch force from 2.93 N to 4.77 N and is adopted.

3.4 In-hand retention during a 180° wrist roll

`exp_hold_test.py` runs the full scene: stick placed at $x_o = 0.075$ m, 0.5 s settle, 1 s close, 0.5 s hold, then Panda joint 7 rolls by 180° in 4 s (palm up $\rightarrow$ sideways $\rightarrow$ down) while the palm tilt, the stick position in the palm frame and the contacts are logged every 10 ms. The stick counts as held while its centre is within 8 cm of the palm centre; the tilt at which it first leaves is the drop tilt. Sliding friction is swept by overwriting the six capsule geoms (priority makes this the effective value), and the cone dimensionality at $\mu = 1.5$.

Table 6 and Fig. 6 give the result. The pinch is friction-limited at closing: for $\mu \leq 0.4$ the stick is squeezed out between the converging thumb pad and index link before the roll starts (the resting surface itself is nearly level, contact normals within 8.4° of vertical, so this is a squeeze-out, not a slide-off), while for $\mu \geq 0.6$ (friction angle 31°) it stays through the whole roll, including hanging palm-down, with 5 mm to 15 mm of slip. The slip is not monotonic in μ because the metric includes the compliant motion of the finger servos dragging the stick as gravity loads change. Disabling torsional and rolling friction (condim 3) lets the stick spin out of the two-point pinch at 58.8°; condim 4 (torsional only) already holds. The scene value $\mu = 1.5$, condim 6 therefore sits with a factor 2.5 of margin above the retention threshold, which is what “relatively sticky, easy to

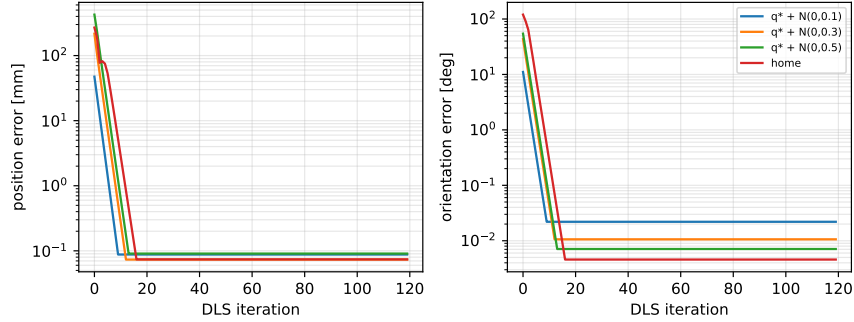


Figure 5: Damped-least-squares convergence to the palm-up target from seeds around the global solution and from the Panda home pose.

Table 5: Grasp sweep in the hand-only scene (0.8s settle, 1s close, 0.8s hold). x_o : stick offset along the fingers from the palm centre; lift: extra clearance applied by the placement routine; n_c , $F_{\max}$: hand-stick contacts and peak normal force after closing.

posture	x_o [m]	lift [mm]	n_c	$F_{\max}$ [N]	links in contact
B	0.035	19	3	0.04	pinky prox., thumb mid., wrist
B	0.055	0	3	0.23	index prox., thumb mid.
B	0.075	0	2	2.93	index prox. abd., thumb pad
B	0.095	0	3	0.02	index prox. abd., wrist
B_curl	0.035	19	4	0.02	pinky prox., thumb mid., wrist
B_curl	0.055	0	3	0.14	index prox., thumb mid.
B_curl	0.075	0	2	3.15	index prox. abd., thumb pad
B_curl	0.095	0	0	0.00	–
B_thumb	0.035	19	3	0.05	pinky prox., thumb mid., wrist
B_thumb	0.055	0	3	0.04	index prox., pinky prox., wrist
B_thumb	0.075	0	2	4.77	index prox. abd., thumb pad
B_thumb	0.095	0	2	0.03	thumb prox. abd., thumb pad

grasp” was taken to mean. No instability warnings occurred in any of the 12 trials.

3.5 Chopstick pinch-lift bench

Whether the *objects* can be picked up with chopsticks is a property of their mass, shape and friction, so it is measured separately from the hand’s skill. In `scenes/chopstick_pinch_test.xml` two chopsticks are welded to mocap bodies (`solref` 4ms) and driven like ideal tweezers by `exp_chopstick_pinch_lift.py`: held tips-down at 20° as a person holds them, they straddle the item along x with 4 mm of clearance, descend from 4 cm above until the tips (the last 2 cm) are at the item’s height, close over 0.5s to a gap of $w - \delta$ (w = item width), hold 0.3s (pinch force), then lift 80 mm in 1s and hold 1s. Success means the item rose more than 60 mm and stayed between the tips. The tilt matters: horizontal sticks at the peanut’s 4.5 mm height caught on the plate rim and never reached it.

Table 7 and Fig. 7 show that a pinch of 0.05 N to 0.2 N, far below the 4.7 N to 5.4 N the hand exerts on a stick, lifts every item. The flat-faced cubes are lifted at every squeeze with either friction, because the two contact normals are antiparallel and the pinch holds geometrically; the round food needs the friction cone and therefore a slightly larger squeeze with plain bamboo; the

Table 6: Hold test. Left: sliding-friction sweep (condim 6). Right: condim sweep at $\mu = 1.5$. n_c , $F_{\max}$ after closing; slip = displacement of the stick in the palm frame relative to the closed state, at the drop or at the end.

μ	n_c	$F_{\max}$ [N]	drop tilt [deg]	slip [mm]		condim	n_c	drop tilt [deg]	slip [mm]
0.1–0.4	0	0.00	lost at close	–					
0.6	2	4.65	held (180)	5.1		3	2	58.8	19.5
0.8	2	5.36	held (180)	9.8		4	2	held (180)	11.9
1.0	2	5.19	held (180)	12.6		6	2	held (180)	12.6
1.5	2	4.87	held (180)	12.6					
2.0	2	4.66	held (180)	14.8					

Table 7: Pinch-lift bench: smallest squeeze δ that lifts each item and the pinch normal force at that squeeze, for the sticky sticks ($\mu = 1.5$) and plain bamboo ($\mu = 0.8$); “–” = not lifted for any $\delta \leq 3$ mm. The cubes are lifted at every δ with either friction; the peanut at $\mu = 1.5$ lifts at 1 mm to 2 mm (0.17 N at 2 mm) and is squeezed out at 3 mm (1.3 N).

item	mass [g]	width w [mm]	$\delta_{\min}$, F at $\mu = 1.5$	$\delta_{\min}$, F at $\mu = 0.8$
tofu cube	8.40	20	0.5 mm, 0.05 N	1.0 mm, 0.10 N
dumpling	8.29	22	0.5 mm, 0.04 N	1.0 mm, 0.08 N
cherry tomato	5.58	22	0.5 mm, 0.04 N	1.0 mm, 0.08 N
peanut	0.67	9	1.0 mm, <0.01 N	–
sugar cube	3.69	16	0.5 mm, 0.05 N	0.5 mm, 0.05 N
pine cube	7.03	25	0.5 mm, 0.05 N	0.5 mm, 0.05 N
foam die	1.88	25	0.5 mm, 0.05 N	0.5 mm, 0.05 N

0.7 g, 9 mm peanut is the hard case: it needs 1 mm to 2 mm of squeeze with the sticky sticks, is squeezed out at 3 mm (1.3 N), and is never lifted with $\mu = 0.8$. The results are what a person experiences with chopsticks, and they confirm that the sticky contact class is what makes the smallest item graspable.

3.6 Two-chopstick pickup by the arm

Table 8 and Fig. 8 give the result of the sequence of Section 2.7 for the six objects of the main scene with the holding pose of Table 3. **The 16 mm sugar cube is picked up:** the sticks stay in the hand through the 6 s approach (axis gap 27 mm before and after), the closing point is placed within 1.4 mm of the cube’s far face, both sticks load the cube (0.09 N peak during the pinch, two contacts), and the supination lift raises it by 71 mm where it stays between the sticks. The run is repeatable (five consecutive runs, 51 mm to 71 mm as the lift height was raised from 30 to 50 mm). The other five objects are not lifted, each for a measured reason: the 25 mm to 28 mm cubes and the dumpling need an opening wider than the 27 mm this hold gives at the tips, and opening the sticks by more than about 35 mm with the fingers releases the sandwich (with a 34 mm opening the upper stick keeps two contacts, at 38 mm it drops); the pinch station further back along the sticks lands 100 mm behind the tips, where the fingers would hit the object; the 22 mm tomato is reached and placed within 1.3 mm, but the sticks close to 17 mm with one contact and slide over the sphere; the tofu on the plate loses a stick during the longer-reach approach.

The ingredients that made the difference, each found by instrumenting a failed run: not pressing the thumb on the upper stick (a press pins it: flexion then moves the tip by 12 mm rad^{-1} instead of 34 mm rad^{-1}); measuring the tip Jacobian instead of assuming that flexion acts along the palm

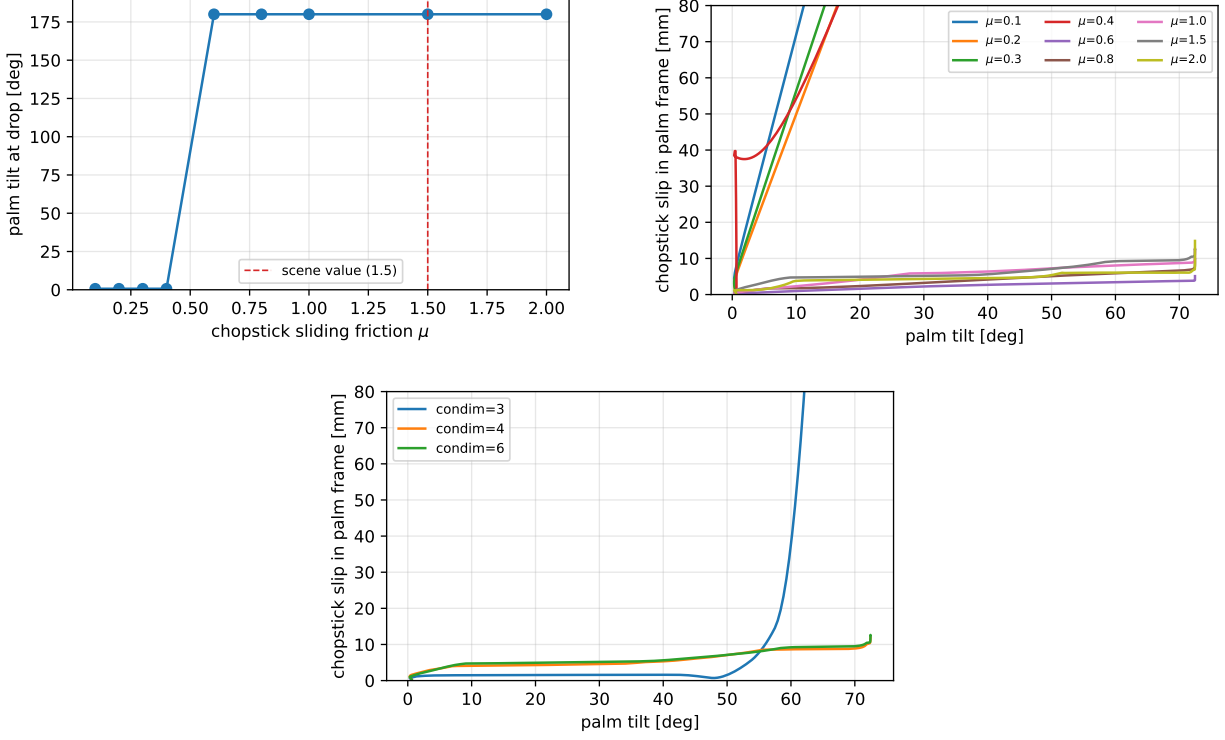


Figure 6: Retention during the 180° roll: drop tilt versus sliding friction (top left; 180° means never dropped, 0.7° means the stick was already gone when the roll started), in-hand slip versus palm tilt for every μ (top right) and for the three contact dimensionalities at $\mu = 1.5$ (bottom).

normal; orienting the hand from the measured closing direction (in this crossed pose it is lateral in the palm, so the palm faces up with the tips level, rather than sideways with the tips offset in height by 30 mm); targeting the lower tip’s height rather than the closing point’s (at a 60° tilt the lower tip was 30 mm lower and dug into the table); placing the static stick at the far face with 3 mm of clearance instead of 1 mm (the arm’s stiffness pressed the cube with 580 N); and stopping the pinch on the moving stick’s own force rather than on any contact. Eleven pick-up variants preceded the working one; all are listed on the project page.

3.7 Tool bench

`exp_tool_bench.py` drives each tool through a mocap weld (an ideal wrist) in `tool_bench_test.xml` (Fig. 9):

- **Scissors.** With the blades horizontal, the rope (segment 5) placed 35 mm from the pivot and 8 mm inside the wedge, and the pivot ramped from 0.6 rad to 0 in 1.5 s, the rope is cut at $t = 0.84$ s, $\theta = 0.28$ rad (Eq. (2) with $x = 35$ mm gives 0.286 rad), and the freed lower piece falls 88 mm to the table; the card (piece 3) is cut at $t = 0.83$ s and its lower half drops 136 mm. The approach must come from the side away from the stand post: coming through it rotated the scissors in their soft weld and forced the pivot past its limit.
- **Tweezers.** Pointing down over an M6 nut with the pads on its flats, closing torques of 0.10, 0.20, 0.30 and 0.40 N m give tip forces of 0.63, 1.54, 2.48 and 3.42 N and the nut is lifted 49 mm every time (it weighs 0.024 N); with the earlier 10 mm rest gap the pads landed on the

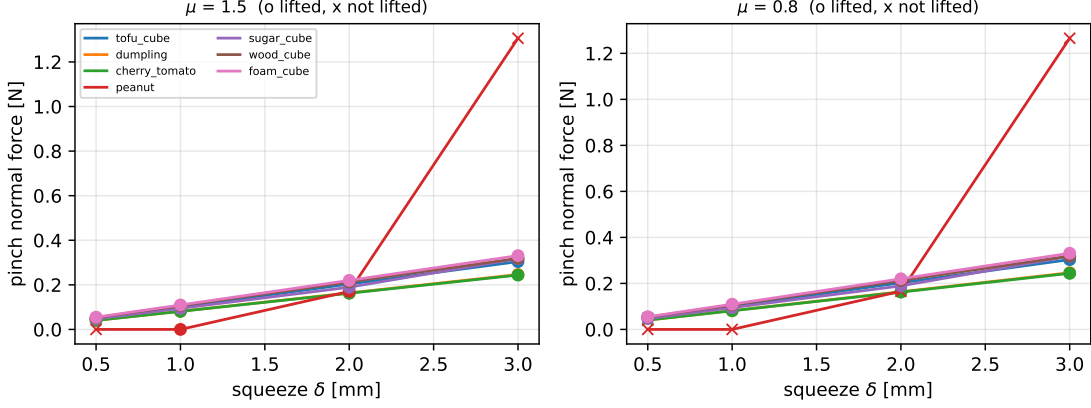


Figure 7: Pinch-lift bench: pinch normal force versus squeeze for every item at $\mu = 1.5$ (left) and $\mu = 0.8$ (right); circles lifted, crosses not lifted.

Table 8: Two-chopstick pickup with the arm and the free sticks: object width, axis gap after the approach, closing-point placement error, axis gap before and after the pinch, chopstick-object contacts and peak normal force during the pinch, object rise after the supination lift.

object	width [mm]	move [mm]	err. [mm]	open [mm]	closed [mm]	n_c	F [N]	rise [mm]	result
sugar cube	16	27	1.4	25	22	2	0.09	+71	lifted
tofu cube	20	440	104.8	473	469	0	0.00	+0	no
cherry tomato	22	31	1.3	32	17	1	0.08	+0	no
dumpling	28	98	28.2	38	38	0	0.00	-0	no
foam die	25	33	25.7	58	56	1	1.50	-0	no
pine cube	25	33	1.4	46	28	1	0.23	-0	no

nut’s corners instead.

- **Pliers.** The jaws (position servo, $k_p = 20$) close on the bolt head with two contacts and a 28 N jaw force and lift the bolt 49 mm. The bolt has to sit 10 mm from the fixed jaw: a pivoting jaw arrives on an arc, and with the item mid-gap the pad came down on the head from above and knocked it away.
- **Drill.** With the bit 1.3 mm from the screw head the coupling is engaged, the trigger is pulled to 30 rad s^{-1} : the chuck runs at 29.3 rad s^{-1} , makes 9.33 turns in 2 s and the screw advances 16.69 mm against 16.80 mm from Eq. (3); the coupling error stays below 0.004 rad.

4 Implementation and Usage

Running. `python scripts/view_scene.py [--scene NAME --key KEY]` opens any scene in the passive viewer (3 toggles collision geometry); `python -m mujoco.viewer --mjcf scenes/<scene>.xml` works as well because every scene is plain MJCF. `scripts/demo_chopsticks_pickup.py --object tofu_cube` replays the two-chopstick pickup interactively.

Package. `wuji_sim.Scene` compiles a scene and exposes `hand` (`WujiHand2Right`: joint/actuator/site indices, `set_target`, `palm_pose`, `fingertip_positions`), `arm` (`PandaArm`: `ik_site` DLS and `ik_site_global`), `free_bodies`, contact queries and offscreen rendering; `wuji_sim.tasks` holds the

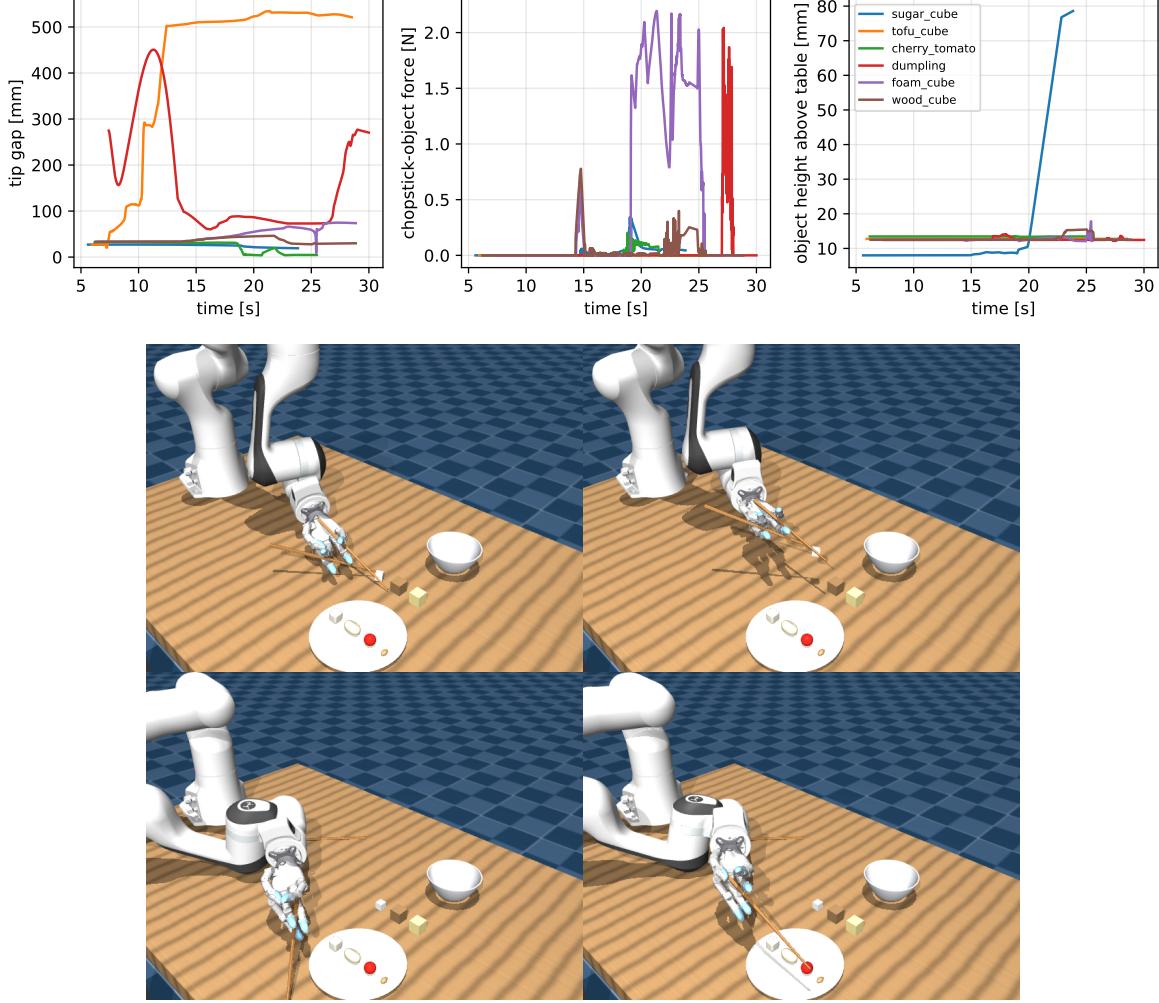


Figure 8: Two-chopstick pickup with free sticks: axis gap, chopstick–object normal force and object height versus time for the six objects (top); the hand after the pinch and after the supination lift for the sugar cube and the tofu (bottom).

primitives used by the experiments (place_chopstick_in_palm, place_body_in_palm, close_grasp, roll_wrist, Cuttable/cut_update, drill_motor_from_trigger); wuji_sim.config is the single home of every shared constant (paths, table height, base pose, palm-up target, grasp posture, object lists and pinch widths).

Regeneration. make_chopstick_mesh.py and make_object_meshes.py → make_collision_meshes.py → convert_upstream_hand.py → build_arm_hand.py → solve_palm_up_pose.py (one keyframe per arm scene) → experiments (settle, hold, pinch-lift, pickup, tool bench) → render_scenes.py; run_all_experiments.py chains them (the CoACD step is skipped by default, its output is committed). make -C docs builds this PDF.

Hardware path. The 20 hand actuators are named after the WUJI SDK convention $r_{\{TH,FF,MF,RF,LF\}J\{0..3\}}$ (J0 flexion, J1 abduction, J2, J3), the joint angles are in radians with the upstream zero and sign conventions, so a joint-space target vector in config.HAND_JOINTS

order maps one-to-one to the real hand; the arm uses the Menagerie Panda joint order. Only `wuji_sim.config`, the scene `<frame>` elements and the keyframe carry scene-specific numbers, so a real-robot driver can replace Scene behind the same `set_target/q` interface. Swapping the hand (the left hand or Beta 1) means pointing `convert_upstream_hand.py` at the other upstream file and rebuilding the composite; adding an object means one more standalone MJCF and one `<frame>` in the scene.

5 Limitations

- **The free-stick grip is a between-fingers variant with a 27 mm opening.** This hand has no thumb web, so the standard grip’s lower-stick anchor does not exist; the sticks are clamped between finger pairs, which fixes their yaw only by friction, and the pose that holds is a crossed (scissors) one: the object is pinched between the upper tip and the lower stick’s shaft. Only the 16 mm sugar cube is lifted; objects wider than about 22 mm need an opening the fingers cannot give without releasing the sandwich, and the pinch does not yet handle a sphere. The constructive tripod grip the user described (index pad on the upper stick, middle finger resting under it, thumb pressing) was attempted (`MODE = "tripod"` in the search script) and no sample passed the geometric stage: the curled ring fingertip sits behind the web point and the index pad does not reach a stick that passes the palm edge. The hand’s actuator gains are scaled (`config.HAND\_GAIN\_SCALE`) to make the clamps hold at all.
- **Tools are operated by the bench, not by the hand.** The tool scenes place each tool under the palm-up hand, and the bench validates the tool physics with mocap welds and scene actuators; grasping a tool with the hand and working it is future work. Cutting is a geometric rule (Eq. (2)) on rigid chains, the card is six rigid pieces, bit engagement is an equality switched on by proximity, and the pliers’ 28 N jaw force is whatever the position servo produces, not a regulated grip.
- **Hand gains are placeholders.** Upstream carries the first-generation gains until Hand 2 system identification is done; the 5°–6° gravity sag and the 4.7 N to 5.4 N pinch force are properties of these gains, not of the hardware.
- **Fingertips are rigid.** The Beta 2 tactile pads are rigid hulls; the soft-pad meshes shipped upstream are not attached, so contact patches are points on convex hulls and the grasp relies on the friction model rather than on pad compliance.
- **Palm geometry is approximate.** Even the CoACD palm deviates up to 5 mm from the mesh on the palm side.
- **Objects are rigid primitives.** Tofu and dumpling do not deform (their compliance is a contact time constant), the collision shapes are boxes, spheres and ellipsoids under the visual meshes, the plate rim is sixteen blocks, and the friction and density values are literature-style estimates, not measurements. The chopstick’s square section is replaced by capsules, so edge contacts are not modelled, and $\mu = 1.5$ is above plain bamboo on purpose; Table 7 quantifies what $\mu = 0.8$ changes. The $1 \times 10^{-6} \text{ kg m}^2$ chopstick armature is a numerical device that overstates the axial inertia by $\sim 50\times$.
- **Home pose points the fingers sideways.** A straight-ahead palm-up pose at table height is at the Panda’s joint limits (Section 2.8).
- **Mount bolt angle assumed.** Eq. (1) assumes the adapter is bolted with its x axis along the flange x ; a rotation by a multiple of 90° changes only the joint-7 value of the keyframe.
- **Arm servo compliance.** The palm centre sags 5.2 mm under the hand’s weight with the Menagerie gains.

- **Throughput.** The full place setting runs at about $2.5\times$ real time single-threaded; the bowl’s 32 wall blocks and the plate rim are the main cost and can be removed for faster training scenes.

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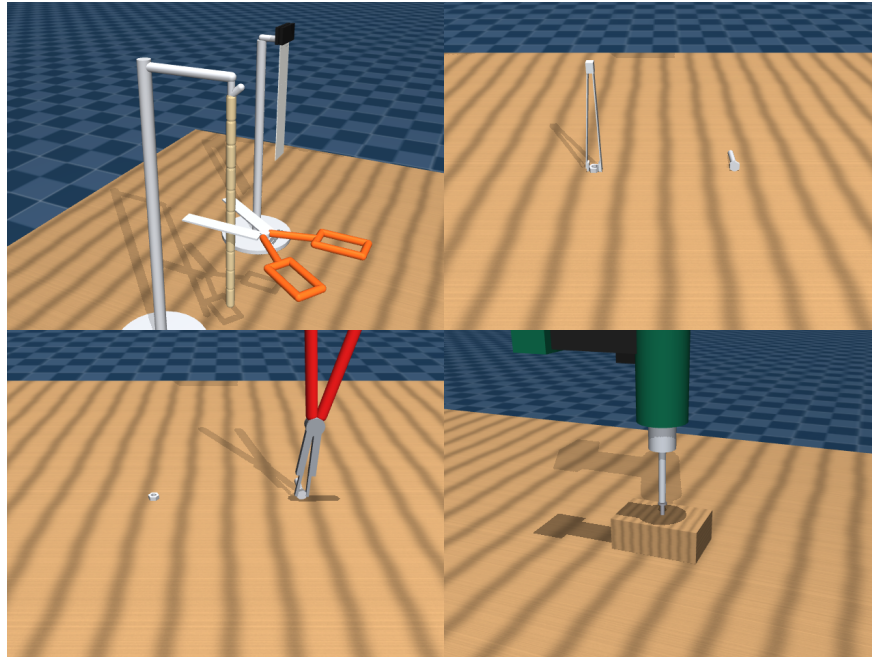
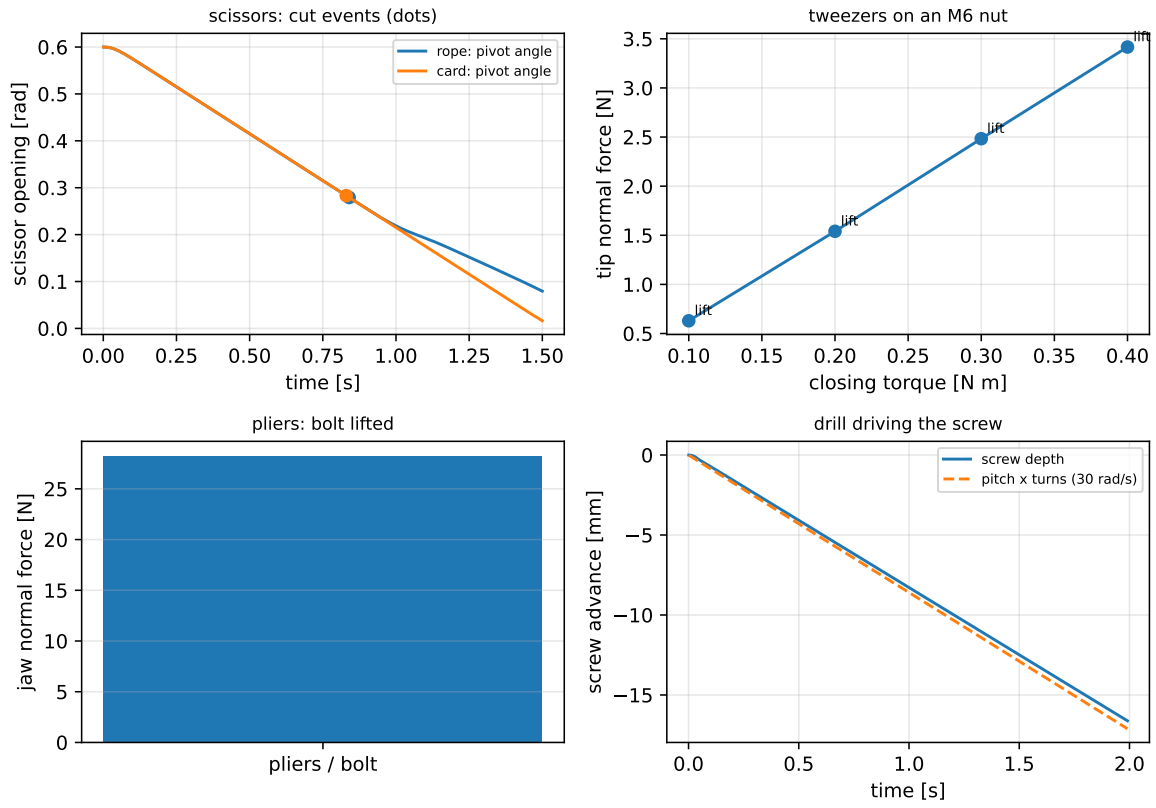


Figure 9: Tool bench: scissor opening angle with the cut events, tweezer tip force versus closing torque, pliers jaw force, and screw advance versus time against the pitch line (top); the four stations at the moment of cutting, pinching and driving (bottom).